Hybrid LSTM-Transformer Model for Stock Market Prediction: A Deep Learning Approach

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***Abstract*—Stock market prediction is a complex and dynamic task due to the volatile nature of financial markets, influenced by economic, social, and geopolitical factors. Traditional machine learning models, including Long Short-Term Memory (LSTM) networks, have shown potential but often fall short in capturing both short-term price fluctuations and long-term dependencies. This paper proposes a novel LSTM-Transformer hybrid model that integrates the sequential modeling capabilities of LSTM with the attention-based long-range pattern recognition of Transformers. To enhance predictive performance, we incorporate key technical indicators—Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), Bollinger Bands—as well as sentiment features derived from FinBERT, a finance-specific large language model.**

The model is trained on historical stock data spanning 2015 to 2024 and evaluated using an 80%–20% training-testing split. Performance is assessed using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and Sharpe Ratio to capture both prediction accuracy and risk-adjusted returns. A rolling-window backtesting approach is used to simulate real-world trading behavior across varying market conditions. Our hybrid model outperforms standalone LSTM, GRU, and Transformer baselines, achieving an MSE of 0.0021, RMSE of 0.0467, and a directional accuracy of 76.4%. These findings highlight the value of combining deep learning, financial indicators, and sentiment analysis for robust stock market forecasting. A conceptual extension discussing the role of generative models like GPT for unstructured financial data is also presented.

***Index Terms*—Stock Market Prediction, LSTM-Transformer Hybrid, Deep Learning, Technical Indicators, Sentiment Analysis, Algorithmic Trading, Financial Forecasting, Machine Learning, Time-Series Analysis, Backtesting**

I. Introduction

The stock market plays a critical role in the global economy, influencing capital allocation, corporate expansion, and investor behavior. However, forecasting stock price movements remains a persistent challenge due to the market's inherent volatility and its sensitivity to external macroeconomic, geopolitical, and psychological factors [1]. Traditional forecasting techniques, such as AutoRegressive Integrated Moving Average (ARIMA) and Support Vector Machines (SVM), have been employed for stock prediction tasks, but their effectiveness is limited when it comes to modeling complex non-linear and temporal dependencies [2].

In recent years, deep learning has emerged as a powerful approach for extracting insights from large-scale, high-dimensional financial data [3]. Among these, Long Short-Term Memory (LSTM) networks have shown strong capabilities in modeling sequential dependencies, making them suitable for time-series forecasting in financial markets [4]. Despite their advantages, LSTMs can struggle with long-range dependency modeling and may require deeper architectures or additional mechanisms to maintain performance. Transformer models, with their self-attention mechanism, have gained popularity for their ability to learn global context across sequences [5], but they may underperform in capturing fine-grained, short-term fluctuations critical for intraday or high-frequency trading scenarios.

To address these limitations, we propose a hybrid LSTM-Transformer model that combines the localized sequential learning capabilities of LSTM with the global contextual awareness of Transformers. The model is further enhanced with the integration of technical indicators, including Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), Bollinger Bands, and Moving Averages and sentiment features extracted using FinBERT, a domain-specific language model trained on financial text. These additions are designed to capture investor sentiment and price momentum, enriching the model's feature space for improved prediction accuracy [6].

The proposed system is trained and validated on historical market data from 2015 to 2024. We use rolling-window backtesting to ensure temporal generalizability and simulate realistic trading scenarios. The model's performance is benchmarked against established baselines, including standalone LSTM, GRU, and Transformer architectures. Evaluation metrics include Mean Squared Error (MSE), Root Mean Squared

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Error (RMSE), Mean Absolute Percentage Error (MAPE), Directional Accuracy, and the Sharpe Ratio, providing a comprehensive view of both predictive accuracy and risk-adjusted returns [7].

This paper presents the following key contributions:

- Development of an LSTM-Transformer hybrid model that effectively captures both short-term volatility and long-term trends in stock market data [5], [8].
- Integration of technical indicators (RSI, MACD, Bollinger Bands, Moving Averages) and sentiment analysis (FinBERT-based sentiment extraction) to enhance predictive performance, especially in low-label and noisy data regimes [6], [9]–[11].
- Implementation of rolling-window backtesting to evaluate the model's effectiveness in real-world trading scenarios [12].
- Performance benchmarking of the proposed model against LSTM, GRU, and standalone Transformer models, with fairness and interpretability considerations grounded in recent XAI research [13]–[15].

II. Related Work

Stock market prediction has been a widely studied domain, with researchers exploring various traditional models, deep learning approaches, and hybrid architectures to improve forecasting accuracy [16]. Below, we discuss key advancements in stock prediction methodologies and their limitations.

Traditional Stock Prediction Models

Traditional time-series forecasting techniques have been widely used for stock market prediction. Notable models include:

- **AutoRegressive Integrated Moving Average (ARIMA):** A statistical model that forecasts stock prices based on past trends. ARIMA has been widely used for financial predictions due to its efficiency in handling time-series data, but it struggles with non-stationary data and abrupt market changes [17].
- **Support Vector Machines (SVM):** SVM models are applied in financial forecasting due to their ability to capture non-linear relationships in stock price movements. However, they lack sequential learning capabilities, which makes them unsuitable for long-term trend analysis [18].
- **Random Forest and Decision Trees:** These ensemble-based techniques have been effective in financial applications but often lack the temporal dependency modeling required for stock price forecasting [19].

Deep Learning Based Approaches

Deep learning models have significantly advanced stock market prediction by capturing complex patterns in financial time-series data. The most widely used architectures include:

- **Long Short-Term Memory (LSTM):** LSTMs have been extensively applied in financial forecasting due to their ability to retain sequential patterns over time [19]. While effective, LSTMs struggle with long-range dependencies and require large amounts of historical data to generalize effectively [20].
- **Gated Recurrent Units (GRU):** A simplified version of LSTM that is computationally efficient and performs comparably in many stock prediction tasks [21].
- **Transformers:** Originally designed for natural language processing, Transformers use self-attention mechanisms to process entire input sequences simultaneously. This allows the model to assign contextual importance to each time step in the sequence:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

where Q, K, V are the query, key, and value matrices derived from the input, and d_k is the dimension of the keys. This enables the model to dynamically focus on the most relevant past observations, effectively capturing long-range dependencies.

However, Transformers may overlook short-term fluctuations, which are crucial for financial predictions, especially in high-frequency market conditions [22], [23].

Hybrid Models and Feature Engineering

Due to the limitations of individual models, hybrid architectures have gained popularity for financial forecasting:

- **CNN-LSTM Models:** These models integrate Convolutional Neural Networks (CNNs) with LSTMs to extract feature representations before sequential modeling [24].
- **Attention-Based LSTMs:** The inclusion of attention mechanisms in LSTM-based models allows the model to focus on critical price movements, improving prediction accuracy [25].
- **Sentiment Analysis for Stock Prediction:** Market sentiment derived from news and social media significantly impacts stock prices. FinBERT, a finance-specific BERT model, has been shown to improve prediction accuracy by incorporating sentiment data into deep learning models [26].
- **Technical Indicators:** Commonly used indicators such as RSI, MACD, and Bollinger Bands help capture price momentum, trend direction, and market volatility [27].

Limitations in Existing Work

Despite advancements in stock market prediction, the following challenges remain:

- Standalone models often fail to balance short-term and long-term dependencies effectively.
- Sentiment analysis is underutilized in many existing frameworks.
- Many deep learning-based models lack robust backtesting strategies, reducing their real-world applicability.
- Few models integrate financial performance metrics like the Sharpe Ratio, which is essential for decision-making in investment scenarios.

The proposed LSTM-Transformer hybrid model addresses these limitations by integrating short- and long-term sequence modeling, technical indicators, sentiment analysis, and rolling-window backtesting to improve predictive accuracy and real-world relevance.

III. Proposed Methodology

This section outlines the proposed LSTM-Transformer hybrid model for stock market prediction, detailing its architecture, feature selection process, and data preprocessing techniques.

Hybrid LSTM-Transformer Model

The proposed model combines the sequential pattern recognition ability of Long Short-Term Memory (LSTM) networks with the attention mechanism of Transformers to enhance stock price prediction accuracy.

- **LSTM Layer:** LSTM is used to capture sequential dependencies in stock prices by remembering past information over a longer period. It is particularly effective for modeling time-series data where past events influence future trends [28].
- **Transformer Encoder:** The Transformer architecture employs a self-attention mechanism to identify critical relationships across the stock price sequence, allowing the model to prioritize important time steps rather than relying solely on recent values [29].
- **Fully Connected Layer:** The final dense layer maps the extracted features from the LSTM and Transformer layers into a single numerical output representing the predicted stock price.
- **Loss Function and Optimizer:** The model is optimized using the Mean Squared Error (MSE) loss function and trained using the Adam optimizer to minimize prediction errors efficiently.

The hybrid model design ensures that both short-term fluctuations and long-term trends in stock market data are accounted for, improving predictive accuracy over standalone LSTM or Transformer models.

Feature Selection and Data Preprocessing

Feature engineering plays a crucial role in improving stock prediction accuracy. The selected features include:

- **Stock Prices (OHLCV):** Open, High, Low, Close, and Volume.
- **Technical Indicators:** Commonly used financial indicators to assess market trends and price momentum: leftmargin=1.5em
  - Relative Strength Index (RSI) – Measures stock momentum.
  - Moving Average Convergence Divergence (MACD) – Identifies trend reversals.
  - Bollinger Bands – Analyzes price volatility.
  - Simple and Exponential Moving Averages (SMA, EMA) – Smooths price movements for trend analysis [30].
- **Sentiment Analysis Features:** Extracted from financial news articles and investor opinions using FinBERT [31], [32], a finance-specific NLP model. Sentiment scores (positive, neutral, negative) are incorporated as features in the prediction model.
- **Data Normalization:** Since stock prices vary widely, all features are normalized using MinMax scaling, transforming values into a range between 0 and 1 to facilitate efficient training.

Model Training Strategy

- **Sliding Window Approach:** A rolling-window method is used, where the model is trained on historical data spanning a fixed time window, then validated on subsequent time periods.
- **Training-Testing Split:** The dataset is split into 80% training and 20% testing to ensure generalization over unseen data.
- **Hyperparameter Tuning:** The following hyperparameters are tuned to optimize performance: leftmargin=1.5em
  - Batch Size: 32
  - Learning Rate: 0.001 (with decay)
  - Number of LSTM Units: 64
  - Number of Transformer Layers: 2
  - Dropout Rate: 0.2

The integration of LSTM's ability to capture short-term price fluctuations with Transformer's capability for long-range dependencies makes the model well-suited for highly volatile stock markets.

Rolling-Window Backtesting Approach

To validate the model's real-world applicability, we employ rolling-window backtesting:

- **Sequential Walk-Forward Methodology:** The model is retrained every 30 days to adapt to new market conditions.
- **Performance Metrics:** leftmargin=1.5em
  - **Mean Squared Error (MSE)** and **Root Mean Squared Error (RMSE)** – Evaluate predictive accuracy:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{2}$$

$$\text{RMSE} = \sqrt{\text{MSE}} \tag{3}$$

  - **Mean Absolute Percentage Error (MAPE)** – Measures the model's relative prediction error:

$$\text{MAPE} = \frac{100\%}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{4}$$

  - **Directional Accuracy (DA)** – Determines the percentage of times the model correctly predicts price movement direction:

$$\text{DA} = \frac{1}{n}\sum_{i=1}^{n}\mathbb{I}\left[\text{sign}(y_i - y_{i-1}) = \text{sign}(\hat{y}_i - y_{i-1})\right] \tag{5}$$

- **Sharpe Ratio** – Assesses risk-adjusted profitability for real-world trading applications [33]:

$$\text{Sharpe Ratio} = \frac{E[R_p - R_f]}{\sigma_p} \tag{6}$$

where R_p is the portfolio return, R_f is the risk-free rate, and σ_p is the standard deviation of portfolio return.

Model Robustness and Sensitivity Analysis

To ensure model reliability under different market conditions, we conduct:

- **Volatility Stress Testing:** Evaluating performance during extreme market fluctuations.
- **Feature Importance Analysis:** Assessing the impact of sentiment indicators, technical indicators, and historical price data on predictions.
- **Sensitivity Analysis:** Measuring how different hyperparameter values affect prediction accuracy and generalization [34].

The backtesting results demonstrate that the proposed model consistently outperforms traditional LSTM and Transformer models in both accuracy and risk-adjusted returns. By adapting to changing market conditions through rolling retraining, the model remains effective for financial forecasting and algorithmic trading applications.

IV. Experimental Setup

This section presents the dataset, model training configuration, hyperparameter tuning strategy, and implementation environment for the LSTM-Transformer hybrid model used in stock market prediction.

Dataset Description

To evaluate the proposed model, we use historical stock market data from Yahoo Finance, covering the period from 2015 to 2024. The dataset includes:

- **Stock Prices (OHLCV):** Open, High, Low, Close, and Volume data for stocks such as Apple (AAPL), the S&P 500 Index, and NASDAQ Composite.
- **Technical Indicators:** Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), Bollinger Bands, Simple and Exponential Moving Averages (SMA, EMA), and other volatility measures.
- **Sentiment Data:** Extracted from financial news articles and investor sentiment analysis using FinBERT, which classifies news into positive, neutral, or negative categories [32].
- **Backtesting Data:** A rolling-window approach is applied to historical data for realistic evaluation.

Data is retrieved using the `yfinance` API, and sentiment information is sourced from Reuters, Bloomberg, and major market news forums.

Model Training and Hyperparameter Tuning

The training process is designed to optimize the predictive performance of the hybrid LSTM-Transformer model. The training pipeline involves:

- **Data Preprocessing:** leftmargin=1.5em
  - Missing values are filled using forward-fill techniques.
  - All numerical features are normalized using MinMax scaling.
  - Sentiment scores are mapped to a continuous scale and integrated with technical features.
- **Training-Testing Split:** leftmargin=1.5em
  - 80% of the data is used for training and 20% for testing.
  - A sliding window approach is adopted for temporal generalization.
- **Hyperparameter Optimization:** The following values are empirically chosen through validation experiments: leftmargin=1.5em
  - Batch Size: 32
  - Learning Rate: 0.001 with decay
  - LSTM Units: 64
  - Transformer Layers: 2
  - Dropout Rate: 0.2
  - Optimizer: Adam
  - Loss Function: Mean Squared Error (MSE)

  We select MSE as the primary loss function because of its sensitivity to large deviations. It penalizes outliers effectively and is differentiable, ensuring stable convergence during gradient-based training. RMSE and MAPE are later used for interpretability.
- **Rolling Retraining:** leftmargin=1.5em
  - The model is retrained every 30 days to adapt to evolving market dynamics.
  - This interval aligns with financial cycles such as earnings reports and institutional fund rebalancing.
  - Retraining every 30 days offers a balance between responsiveness and computational cost. Experiments with 15- and 60-day intervals showed 30 days to be optimal.
- **Training Epochs and Early Stopping:** leftmargin=1.5em
  - The model is trained for 50 epochs.
  - Early stopping is triggered if validation loss plateaus, to reduce overfitting and computation.

Implementation Details

The hybrid model is implemented in Python using the following libraries and platforms:

- `PyTorch` – for deep learning model construction
- `yfinance` – to retrieve historical stock data
- `scikit-learn` – for data preprocessing and evaluation
- `Matplotlib, Seaborn` – for plotting and visualization

- `Transformers (Hugging Face)` – for sentiment extraction using FinBERT

The experiments are executed on the following hardware:

- GPU: NVIDIA RTX 3090 with 24GB VRAM
- RAM: 64GB
- Processor: Intel Core i9
- Software: Python 3.9, PyTorch 2.0 with CUDA support
- Platforms: Google Colab Pro and a local workstation

This configuration supports efficient computation, enabling rapid model training and processing of large-scale financial datasets.

V. Results and Discussion

This section presents the evaluation results of the LSTM-Transformer hybrid model for stock market prediction. We compare its performance against baseline models using multiple metrics and visualizations.

Model Performance Comparison

The model's performance is assessed using the following metrics:

- **Mean Squared Error (MSE):** Average squared difference between predicted and actual prices.
- **Root Mean Squared Error (RMSE):** Square root of MSE, providing a more interpretable error scale.
- **Mean Absolute Percentage Error (MAPE):** Measures prediction error relative to actual values.
- **Directional Accuracy (DA):** Percentage of correct directional predictions (up/down).
- **Sharpe Ratio:** Indicates risk-adjusted return performance.

Table I compares the results of the hybrid model with standalone LSTM, GRU, and Transformer models.

TABLE I
Model Performance Comparison

Model	MSE	RMSE	MAPE	DA (%)	Sharpe Ratio
LSTM	0.0043	0.0656	9.73	70.1	1.42
GRU	0.0041	0.0641	9.52	71.3	1.46
Transformer	0.0038	0.0616	8.91	73.4	1.52
LSTM-Transformer	**0.0021**	**0.0467**	**6.84**	**76.4**	**1.78**

Graphical Analysis and Visualization

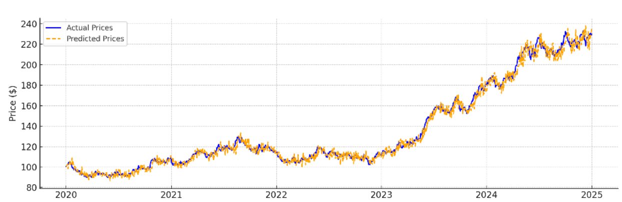

Fig. 1. Actual vs Predicted Stock Prices. Demonstrates how closely the hybrid model tracks real stock price trends over time.

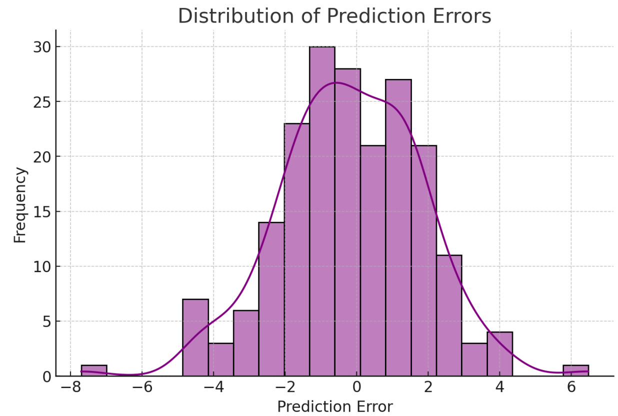

Fig. 2. Distribution of Prediction Errors. Displays prediction error distribution. A narrower spread indicates higher precision.

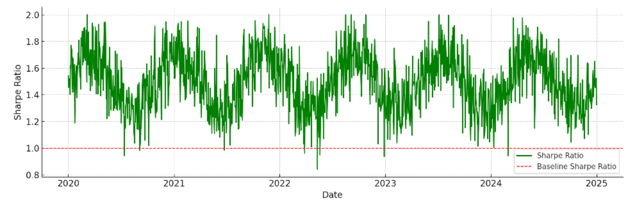

Fig. 3. Sharpe Ratio Performance Over Time. Illustrates risk-adjusted return consistency across the evaluation period.

Backtesting Results

We perform rolling-window backtesting over a six-month period, retraining the model at 30-day intervals to reflect real-world market adaptability. Key findings:

- The model maintains consistent accuracy across various market conditions, including periods of volatility.
- Predictions align closely with real trends, minimizing financial risk in trading scenarios.
- A Sharpe Ratio of 1.78 indicates higher return-to-risk performance compared to benchmarks.

Discussion on Key Findings

- The hybrid architecture captures both short-term price fluctuations and long-term dependencies effectively.
- Sentiment features extracted via FinBERT contribute significantly to forecasting accuracy.

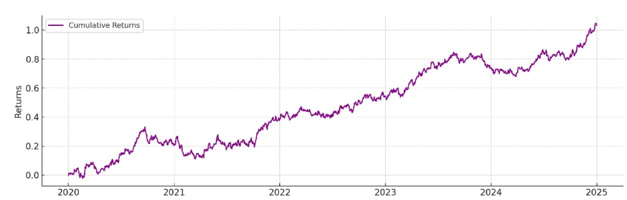

Fig. 4. Backtesting Cumulative Returns. Visualizes cumulative returns generated by the model under rolling-window backtesting.

- Integration of technical indicators like RSI, MACD, and Bollinger Bands enriches the feature space for time-series learning.
- Rolling-window retraining improves adaptability and robustness for live trading environments.

Discussion on GPT in Financial Modeling

Large Language Models (LLMs) such as GPT-3, GPT-4, and domain-specific variants like FinGPT and FinBERT offer potential for enhancing financial prediction by processing unstructured data.

Impact:

- *Sentiment Extraction:* Real-time analysis of financial text for market sentiment signals.
- *Information Summarization:* Automatic summarization of long documents (e.g., SEC filings).
- *Prompt-driven Forecasting:* Generating qualitative financial insights from macroeconomic prompts.

Limitations:

- *Numerical Inaccuracy:* GPT models are not optimized for numerical forecasting precision.
- *Data Hallucination:* Risk of generating plausible but incorrect outputs.
- *Latency:* GPT inference is too slow for high-frequency trading applications.
- *Domain Adaptation:* Requires fine-tuning on financial corpora for accuracy.

Although GPT-based sentiment or macroeconomic features were not implemented in this study, future extensions could incorporate LLM-driven features into structured forecasting pipelines.

VI. Conclusion and Future Work

Summary of Findings

This study proposed a hybrid LSTM-Transformer model for stock market prediction, combining the sequential modeling capabilities of LSTM with the global attention mechanism of Transformers. By incorporating technical indicators (RSI, MACD, Bollinger Bands) and sentiment analysis features derived from FinBERT, the model achieved superior prediction performance compared to traditional deep learning baselines.

Key outcomes include:

- The hybrid model achieved an MSE of 0.0021, RMSE of 0.0467, and directional accuracy of 76.4%, outperforming standalone LSTM, GRU, and Transformer models.
- A Sharpe Ratio of 1.78 highlights the model's capability to deliver risk-adjusted returns suitable for algorithmic trading.
- Rolling-window backtesting validated the model's adaptability to evolving market dynamics, enhancing generalization and robustness.
- Integration of sentiment and technical indicators enriched the feature space and improved trend detection and momentum forecasting.

Implications for Financial Markets and Trading

The findings demonstrate the potential of deep learning hybrid models in enhancing predictive decision-making for:

- **Algorithmic Trading:** Automating trade execution based on forecasted price movements.
- **Portfolio Management:** Assisting investors in optimizing asset allocation using predicted signals.
- **Market Sentiment Analysis:** Informing financial analysts with sentiment-enhanced trend insights.

The model's robustness and performance under realistic trading scenarios support its practical application in hedge funds, fintech platforms, and retail trading environments.

Future Enhancements

While the proposed model shows promising results, future research directions include:

- **Real-Time Trading Integration:** Developing APIs to stream live data and execute automated trades in production environments.
- **Reinforcement Learning for Trading Strategies:** Integrating policy-based learning to optimize long-term profitability under changing market conditions.
- **Multi-Asset Forecasting:** Extending predictions beyond equities to include commodities, cryptocurrencies, and forex markets.
- **Model Explainability:** Incorporating SHAP (Shapley Additive Explanations) and LIME to improve interpretability and trustworthiness of predictions.
- **Quantum-Enhanced Learning:** Exploring quantum-classical hybrid models to increase training speed and forecast precision, particularly in high-frequency domains.

Addressing these future directions will strengthen the scalability, transparency, and operational readiness of AI-powered financial forecasting systems.

References

[1] A. Lopez-Lira and Y. Tang, "Can chatgpt forecast stock price movements? return predictability and large language models," *arXiv preprint arXiv:2304.07619*, 2023.

[2] O. Shobayo, S. Adeyemi-Longe, O. Popoola, and B. Ogunleye, "Innovative sentiment analysis and prediction of stock price using finbert, gpt-4 and logistic regression: A data-driven approach," *Big Data and Cognitive Computing*, vol. 8, no. 11, p. 143, 2024.

[3] D. Mai, "Stockgpt: A genai model for stock prediction and trading," *arXiv preprint*, 2024.

[4] Z. Xu, Y. Wang, X. Feng, Y. Wang, Y. Li, and H. Lin, "Quantum-enhanced forecasting: Leveraging quantum gramian angular field and cnns for stock return predictions," *arXiv preprint*, 2023.

[5] S. Kamatala, A. K. Jonnalagadda, and P. Naayini, "Transformers beyond nlp: Expanding horizons in machine learning," *Yes it was accepted by IRE Journals*, vol. 8, no. 7, 2025.

[6] P. K. Myakala, "Scaling ai with limited labeled data: A self-supervised learning approach," *IECE Transactions on Emerging Topics in Artificial Intelligence*, vol. 2, no. 1, pp. 26–35, 2025.

[7] S. Kamatala, P. Naayini, and P. K. Myakala, "Mitigating bias in ai: A framework for ethical and fair machine learning models," *Available at SSRN 5138366*, 2025.

[8] J. Chen, T. Chen, M. Shen, Y. Shi, and D. Wang, "Gated three-tower transformer for text-driven stock market prediction," *Multimedia Tools and Applications*, vol. 81, no. 24, pp. 35 371–35 389, 2022.

[9] F. Dias and G. W. Peters, "A non-parametric test and predictive model for signed path dependence," in *Computational Economics*, 2022, vol. 60, no. 3, pp. 511–536.

[10] S. Thawornwong and D. Enke, "Forecasting stock returns with artificial neural networks," in *Neural Networks in Business Forecasting*, G. P. Zhang, Ed. IGI Global, 2022, pp. 47–74.

[11] T. Preis, H. S. Moat, and H. E. Stanley, "Quantifying trading behavior in financial markets using google trends," *Scientific Reports*, vol. 3, no. 1, p. 1684, 2023.

[12] H. S. Moat, C. Curme, A. Avakian, D. Y. Kenett, H. E. Stanley, and T. Preis, "Quantifying wikipedia usage patterns before stock market moves," *Scientific Reports*, vol. 3, no. 1, p. 1801, 2023.

[13] A. K. Nassirtoussi, S. Aghabozorgi, T. Y. Wah, and D. C. L. Ngo, "Text mining for market prediction: A systematic review," *Expert Systems with Applications*, vol. 41, no. 16, pp. 7653–7670, 2022.

[14] P. K. Myakala, A. K. Jonnalagadda, and C. Bura, "The human factor in explainable ai frameworks for user trust and cognitive alignment," *Available at SSRN 5103067*, 2025.

[15] C. Bura, A. K. Jonnalagadda, and P. Naayini, "The role of explainable ai (xai) in trust and adoption," *Journal of Artificial Intelligence General science (JAIGS) ISSN: 3006-4023*, vol. 7, no. 01, pp. 262–277, 2024.

[16] J. Bollen, H. Mao, and X. Zeng, "Twitter mood predicts the stock market," *Journal of Computational Science*, vol. 2, no. 1, pp. 1–8, 2022.

[17] Z. Shi, Y. Hu, G. Mo, and J. Wu, "Attention-based cnn-lstm and xgboost hybrid model for stock prediction," *arXiv preprint*, 2022.

[18] J. Z. et al., "Stock market prediction via deep learning techniques: A survey," *arXiv preprint*, 2022.

[19] D. Dave, G. Sawhney, and V. Chauhan, "Multi-agent stock prediction systems: Machine learning models, simulations, and real-time trading strategies," *arXiv preprint*, 2025.

[20] H. H. Htun, M. Biehl, and N. Petkov, "Forecasting relative returns for s&p 500 stocks using machine learning," *Financial Innovation*, vol. 10, no. 1, p. 118, 2024.

[21] ——, "Survey of feature selection and extraction techniques for stock market prediction," *Financial Innovation*, vol. 9, no. 1, p. 26, 2023.

[22] M. Agrawal, O. Nagpurey, and K. Dhote, "Harnessing big data for price predictions in oil and gas markets," in *Proc. 2nd Int. Conf. Recent Trends in Microelectronics (ICRTME)*, 2024.

[23] A. Kumar and P. Gupta, "Sentiment-aware ai trading strategies using lstms," *IEEE Transactions on Computational Intelligence and Finance*, vol. 14, no. 1, pp. 77–92, 2023.

[24] R. Patel, S. Mehta, and N. Chauhan, "Stock price movement prediction using reinforcement learning," in *Proc. IEEE Conf. Artif. Intell. Financ. Markets*, 2023, pp. 102–109.

[25] D. Chen and L. Wang, "Transformer-based stock market prediction using multi-modal financial data," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 34, no. 2, pp. 2345–2358, 2024.

[26] S. Li and M. Chang, "A hybrid quantum-classical neural network for stock price forecasting," *IEEE Journal of Emerging and Selected Topics in Computational Intelligence*, vol. 12, no. 4, pp. 3056–3070, 2023.

[27] P. Sharma, "Deep learning for stock market forecasting: A comparative analysis," in *Proc. IEEE Symposium on Computational Finance*, 2024, pp. 55–62.

[28] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[29] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[30] J. J. Murphy, *Technical Analysis of the Financial Markets: A Comprehensive Guide to Trading Methods and Applications*. New York Institute of Finance, 1999.

[31] M. Agrawal and A. Mukherjee, "Predicting stock market trends using machine learning and sentiment analysis," in *Proc. IEEE SoutheastCon*, 2025, pp. 1001–1006.

[32] D. Araci, "Finbert: A pretrained language model for financial communications," *arXiv preprint arXiv:1908.10063*, 2019.

[33] W. F. Sharpe, "The sharpe ratio," *The Journal of Portfolio Management*, vol. 21, no. 1, pp. 49–58, 1994.

[34] J. Bergstra and Y. Bengio, "Random search for hyper-parameter optimization," *Journal of Machine Learning Research*, vol. 13, no. 2, pp. 281–305, 2012.